Multi-Model Machine Learning Approaches for Fault and Anomaly Detection in Petroleum Product Distribution: A Comparison

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***Abstract*—Maintaining the integrity of petroleum transportation is essential for operational safety and efficiency. Substantial issues arise from faults in fuel transportation systems, including distribution valve malfunctions and anomalies in transit patterns, such as theft or unexpected fuel losses. This work presents a machine learning approach that combines supervised fault detection models, like Logistic Regression, SVM, Random Forest, and XGBoost, with unsupervised anomaly detection methods, such as k-means, Isolation Forest, and Autoencoders. The study employs real fuel transportation data and anomaly detection datasets obtained from petroleum distribution networks. Experimental results show that combining supervised and unsupervised methods leads to better detection accuracy and reliability, making the petroleum delivery system more dependable.**

***Index Terms*—Fault detection, anomaly detection, petroleum transportation, machine learning, predictive maintenance.**

I. Introduction

The petroleum sector is a system that needs to have a constant fuel supply. However, several problems related to the operation of the company may negatively affect efficiency, safety, and profitability. Deficiencies in fuel transportation systems lead to fuel mixing, pollution and financial losses. Environmental pollution can occur by leakages, spills and improper handling of the product during transportation leading to contamination of soil and water. These risks pose significant ecological and economic concerns, hence effective risk assessment and mitigation techniques are required [1]. Operational anormalities in distribution networks include unexpected fuel losses, fraudulent activities, and route variations, suggesting possible inefficiencies or security breaches. A significant vulnerability in fuel distribution networks results from malfunctions in valve devices that divide various fuel types in transport trailers. Defects in these systems may lead to the wrong type of fuel being used in vehicles, which leads to poor fuel quality, reduced vehicle performance, and higher costs to both the distributors and consumers of the fuel [2]. In addition, other operational irregularities, including, unforeseen fuel losses, fraudulent activities, and unauthorized route diversion may point to poor management, security breaches or even fraud within the value chain. Some examples of operational irregularities include: fuel losses, fraud, and changes in the channel, which are unexpected fuel losses, fraudulent actions, and changes in the route. These anomalies may be an indication of poor management or security breaches.

Research has investigated various methods to enhance fuel distribution efficiency to address current challenges. Metaheuristic approaches have been applied to fuel distribution networks for vehicle routing optimization under capacity constraints according to researches [3]. As these methods boost logistical performance they are generally created for route optimization and scheduling needs rather than the identification of mechanical problems or system issues. The research delivers a multi-model machine learning framework that integrates both supervised models to detect faults and unsupervised models to detect anomalies in petroleum distribution to resolve these challenges. The main goal is to boost operational safety, enhance maintenance strategies and complement traditional fuel routing optimization methods.

Despite the fact that these conventional approaches are able to identify obvious errors, they are not effective in identifying complex and evolving patterns that, if ignored, could lead to severe challenges. Because of this limitation, the diagnosis of a malfunction is usually delayed, which results in prolong periods of downtime and increased operational hazards.

This research examines various machine learning methods for defect detection and anomaly recognition in petroleum distribution systems. It integrates supervised and unsupervised

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learning methodologies to deliver a comprehensive analysis of fault patterns and operational anomalies rather than concentrating entirely on route optimization or individual fault detection models. The research examines the correlation between feature engineering and model performance, and how domain specific characteristics can enhance predictive accuracy and minimize false positives

The structure of this paper is as follows: Section II presents a review of fault detection and anomaly detection techniques in industrial and fuel distribution systems, with a focus on machine learning-based approaches. Section III outlines the methodology, including a description of the dataset, preprocessing steps, and details on the supervised and unsupervised models employed for fault detection and anomaly identification. Section IV discusses challenges encountered during the study and talk about the results and evaluates model performance using classification metrics such as accuracy, precision, recall, and F1-score. Section V concludes the paper by summarizing key contributions and highlighting potential areas for future work.

II. Related Works

Most of the papers written related to fault detection and anomaly detection refer to data-driven machine learning models compared to the traditional statistical and rule-based approaches used to spot real-time data anomalies and faulty behaviors during the distribution of petroleum products.

A. Fault Detection in Industrial Systems

The application of machine learning (ML) for fault detection in industrial systems is widespread to predict when maintenance is required and to prevent unexpected system failures. The application of data-driven machine learning models has been found to be more reliable than traditional rule-based methods, especially when operating under changing conditions.

Patil et al. [4] state that the use of intelligent sensors c enable real-time fault detection through ongoing monitoring of system performance. However, there are still issues with the development of models that are able to perform well across a range of operating conditions. In addition, Bode et al. [5] explore the use of machine learning techniques for fault detection in energy systems and highlight the problems of poor feature selection, limited labelled data and generic models to enhance fault classification accuracy.

Vaish et al. [6] present an overview of the use of ML in power system fault diagnosis and demonstrate how supervised and unsupervised learning can enhance fault detection, classification and location. They also highlight the advantages of real time data processing with the help of advanced monitoring tools like PMUs to enhance the reliability of the system.

Nevertheless, there are problems regarding generalization, data quality and robustness that are still relevant for ML models. Future work should be directed at enhancing the adaptability of the model and improving predictive maintenance strategies across various industries.

A popular method for identifying faults in industrial systems, machine learning (ML) ensures predictive maintenance and reduces unplanned failures. Since unpredictable operating conditions are frequently a problem for traditional rule-based approaches, data-driven machine learning models offer a more dependable alternative.

B. Anomaly Detection in Fuel Distribution

Recent research has focused on machine learning techniques for anomaly detection in petroleum transportation, explore both supervised and unsupervised learning approaches to detect network and operational anomalies.

Savic et al. [7] employed Variational Autoencoders (VAE) and KMeans clustering to detect anomalies in cellular IoT used for freight tracking, particularly for perishable and hazardous materials.

Haji et al. [8] utilized KNN, Local Outlier Factor (LOF), and SVM to identify physical anomalies in the transport of petroleum and hazardous gases, emphasizing deviations in parameters like pressure and temperature.

Hindy et al. [9] applied Random Forest and Gradient Boosting (XGBoost) to detect network anomalies in IoT-based logistics systems.

Salej & Trebar [10] analyzed Logistic Regression, SVM, and Gradient Boosting for identifying operational anomalies in petroleum and gas transport, focusing on the challenges posed by edge computing constraints.

Other studies explored hybrid approaches, such as combining Autoencoders, KMeans, and Isolation Forest, to handle real-time data anomalies in IoT-based transportation and hazardous materials transportation.

III. Methodology

This section examines the use of different supervised and unsupervised learning algorithms to detect abnormalities and faults in industrial systems related to petroleum product distribution in different locations. It offers a thorough explanation of datasets, data engineering, preprocessing steps, and algorithm-specific parameter tweaking.

A. Dataset and Preprocessing

Two separate datasets are used: fuel transportation logs and detailed operational records for each affected truck, covering data for normal operating days and mixing event days. This historical viewpoint allows for a more thorough analysis of pre-failure conditions, helping to identify early warning signals that precede mixing events. Petroleum distribution anomaly dataset, which captures irregular transportation patterns and unexpected losses, contains 214,003 instances, with 32,840 corresponding to probable anomalies. This collection mostly covers historical transport data, with the majority of records dating from 2021 to 2024 and recording abnormalities observed during fossil fuel distribution, mainly petroleum products. The dataset also comprises signals collected from various sensors that monitor traffic activity.

The combination of these datasets offers a more robust approach to recognizing and addressing abnormalities in the

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fuel distribution process, resulting in improved accuracy and operational reliability.

B. Preprocessing Steps

A variety of preprocessing methods were applied to the dataset to preserve data integrity and improve model performance. Depending on the properties of the missing attributes, mean or mode imputation was used to handle missing values. Categorical features, such as location and fuel type, were converted using one-hot encoding for compatibility with machine learning models.

Feature engineering played a key role in improving fault detection by identifying fuel loss, unusual transit patterns, and temperature-pressure changes. These features helped the model better recognize faults and anomalies. Standardizing numerical values ensured consistency and reduced bias in models sensitive to scale. By enhancing data quality and highlighting important patterns, these preprocessing steps improved model accuracy and reliability.

Data is split 80-20% into training and testing sets to assess model performance on unseen data. Stratified k-Fold Cross-Validation is applied to prevent overfitting and fine-tune hyperparameters. Additional features, such as aggregated event logs and geospatial attributes, enhance anomaly detection accuracy.

Logarithmic scaling and polynomial feature transformation techniques are used to enhance feature relationships for the models.

A pipeline framework is implemented to automate preprocessing and ensure the same transformations are consistently applied to both training and test datasets. This pipeline, depicted in Fig. 1, ensures data consistency, reproducibility, and scalability.

C. Supervised Models for Fault Detection

1) Logistic Regression: Logistic Regression is a statistical model for classification that predicts the likelihood of an event happening based on past data. It uses the sigmoid function to convert input features into a probability score, making it ideal for binary classification tasks like fault detection. The likelihood of a sample being part of the defective class is expressed as:

$$P(y=1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_n x_n)}} \tag{1}$$

where β_0 is the intercept, β_i are the feature coefficients, and x_i represents the input features. The decision boundary is determined by:

$$y = \begin{cases} 1, & P(y=1|x) \geq 0.5 \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

Although it is simple, Logistic Regression is effective when linear separability exists, but it has difficulties with non-linear fault patterns.

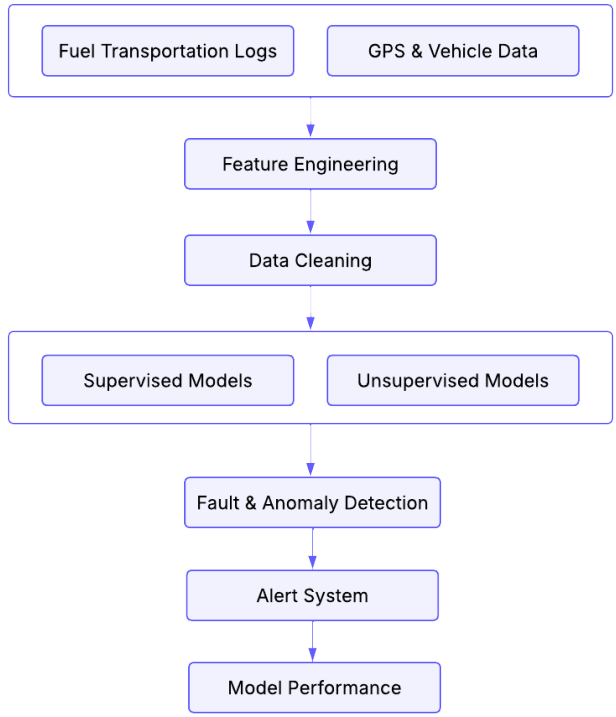

Fig. 1. System's architecture

2) Support Vector Machines (SVM): SVM is a supervised learning algorithm that constructs an optimal decision boundary to separate faulty and non-faulty instances. It is based on the principle of structural risk minimization of statistical learning theory. The optimal hyperplane is given by:

$$w^T x + b = 0 \tag{3}$$

where w is the weight vector, x represents the feature vector and b is the bias term. The objective of SVM is to maximize the margin between support vectors while minimizing classification errors, which is formulated as follows:

$$\min_{w,b} \frac{1}{2}||w||^2 \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1, \forall i \tag{4}$$

SVM's ability to handle high-dimensional feature spaces makes it suitable for fault detection, especially when using kernel functions to identify non-linear connections.

3) Random Forest: Random Forest is an ensemble learning method that enhances classification reliability by merging several decision trees. Every tree is trained on a random portion of the data through bootstrap aggregation (bagging). The ultimate classification result is decided by majority voting

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_T(x)\} \tag{5}$$

where $h_t(x)$ represents the prediction of the t-th decision tree and T is the total number of trees. Random Forest is particularly effective in fault detection scenarios due to its

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ability to handle missing data, reduce overfitting, and improve generalization.

4) XGBoost: XGBoost (Extreme Gradient Boosting) is a refined boosting algorithm intended for the classification of structured data. In contrast to Random Forest, which constructs trees independently, XGBoost creates trees in a sequence, correcting the mistakes from earlier iterations. The objective function consists of a loss function L and a regularization term Ω:

$$\mathcal{L}(\Theta) = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{6}$$

where y_i is the actual label, $\hat{y}_i$ is the predicted label and f_k represents the k-th tree. The regularization term Ω helps control model complexity and prevents overfitting. XGBoost is particularly effective in handling imbalanced datasets and complex fault detection tasks, making it a preferred choice in predictive maintenance applications.

D. Unsupervised and semi-supervised Models for Anomaly Detection

1) k-Means: partitions the data into k groups based on similarity and classifies the data into normal and anomalous patterns by measuring the distance of each data point from its assigned cluster centroid [13] [12]. The most common metric used to calculate the distance between a data point and its centroid is the Euclidean distance:

$$D = \sqrt{\sum_{i=1}^{n} (x_i - c_i)^2}$$

where x_i represents the features of a data point, and c_i is the centroid of the cluster. Data points that are farther from their centroids are considered anomalies.

2) Isolation Forest: is a tree-based method that isolates anomalies by creating partitions based on randomly chosen features and split values [13]. The anomaly score is determined by how quickly a data point is isolated, with anomalies being isolated at a shallow depth in the tree. The score $s(x)$ is calculated as:

$$s(x) = 2^{\left(\frac{E(h(x))}{c(n)}\right)}$$

where $E(h(x))$ is the average path length of a point x across all trees, and $c(n)$ is a normalization constant that accounts for the expected path length in a random partition.

3) Autoencoders: neural networks trained to recognize normal patterns and detect deviations are trained on normal data, and anomalies are detected by the reconstruction error. Points that are not well reconstructed exhibit high error, indicating that they are different from the normal patterns learned by the network [14] [15]. The Mean Squared Error (MSE) is commonly used to compute the reconstruction error:

$$\text{MSE}(x) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{x}_i)^2$$

where x_i is the original data point, and $\hat{x}_i$ is its reconstructed version.

4) KNN and LOF: K-nearest neighbors (KNN) identify global outliers, while the Local Outlier Factor (LOF) detects local anomalies. Their combination improves the detection of anomalies, as suggested in [7] KNN detects anomalies by measuring the average distance from the nearest neighbors k:

$$D_k(x) = \frac{1}{k} \sum_{i=1}^{k} d(x, x_i) \tag{7}$$

where $d(x, x_i)$ is the Euclidean distance between x and its neighbor x_i. Higher $D_k(x)$ values indicate potential anomalies.

LOF evaluates local density deviations. The local reachability density (LRD) is:

$$LRD_k(x) = \frac{k}{\sum_{i=1}^{k} \max(d(x, x_i), RD_k(x_i))} \tag{8}$$

where reachability distance is:

$$RD_k(x_i) = \max(D_k(x), d(x, x_i)) \tag{9}$$

LOF is then computed as:

$$LOF_k(x) = \frac{\sum_{i=1}^{k} \frac{LRD_k(x_i)}{LRD_k(x)}}{k} \tag{10}$$

A higher LOF indicates a stronger likelihood of an anomaly.

IV. Results and Discussion

The results presented hereafter show a comparative analysis of various machine-learning models for anomaly detection and fault analysis in petroleum distribution. The main metrics used to define the model's efficiency and performance are accuracy, precision, recall, f1 score, and roc area under the curve, where it is noticed that both supervised and unsupervised models showed an average to high accuracy in detecting anomalous and faulty cases.

The ROC AUC score, which measures the model's ability to distinguish between normal and anomalous cases, varied across different models, with the highest value observed for KMeans & Isolation Forest (0.93) and XGBoost (0.90), indicating strong predictive power. Models like SVM (0.90) and Random Forest (0.85) also demonstrated high discriminative ability, while Logistic Regression (0.68) and VAE (0.73) showed lower effectiveness in distinguishing anomalies.

Recall, a crucial metric in anomaly detection, which measures the ability of a model to correctly identify positive instances out of all actual positives, was highest for KMeans & IF (0.89) and XGBoost (0.80), ensuring that most anomalies were successfully identified. However, Logistic Regression (0.40) and SVM (0.40) had low recall, meaning many anomalies were missed despite their high precision. This trade-off highlights that models prioritizing high recall—such as XGBoost and KNN & LOF (0.83)—are better suited for minimizing false negatives, which is critical in fault detection.

Precision, a metric that measures how many detected anomalies were actual anomalies, was highest for Logistic

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Regression (0.83) and SVM (0.83), suggesting they were conservative in flagging anomalies but at the cost of missing several. On the other hand, VAE (0.37) had the lowest precision, indicating that many of its anomaly predictions were false positives.

F1 Score, which balances precision and recall and indicates a good trade-off between correctly spotting anomalies (recall) and minimizing false positives, favored KMeans & IF (0.84) as the most well-rounded model, followed by XGBoost (0.70) and KNN & LOF (0.70).

In summary, KMeans & Isolation Forest emerged as the best-performing model, achieving the highest accuracy (0.95), recall (0.89), and ROC AUC score (0.93), making it the most reliable for anomaly detection in petroleum distribution.

The KMeans and Isolation Forest hybrid approach are the most effective in anomaly detection due to their complementary strengths. KMeans groups normal transportation patterns, allowing for clear distinction between standard operations and anomalies. It segments data into well-defined clusters, making deviations easily identifiable. Isolation Forest, on the other hand, excels at detecting rare and unexpected deviations by isolating anomalies quickly. This combination makes it particularly suitable for identifying complex distribution irregularities in petroleum distribution. The hybrid model can handle high-dimensional data effectively, maintaining accuracy even when dealing with multiple factors influencing transportation patterns. It also offers scalability and flexibility, adapting to changes in operational norms and working with unlabelled data. The combination of KMeans and Isolation Forest ensures generalization across datasets and high sensitivity to rare events, enhancing anomaly detection performance.

XGBoost and KNN & LOF also performed well, offering a good trade-off between recall and precision, while Logistic Regression and VAE struggled with recall and precision balance, making them less effective for anomaly detection.

A. Limitations and Challenges

This study also revealed certain limitations. The relatively small dataset size constrained the models' ability to generalize, especially in complex scenarios. Increasing the amount of training data, particularly with more examples of faulty instances, could enhance performance. Additionally, anomalies in fuel transportation are highly context-dependent; factors such as delivery importance and cargo urgency play a crucial role in determining what qualifies as an anomaly. Automated systems often lack this contextual understanding, making it difficult to distinguish between normal and abnormal behaviors.

Another significant challenge was the lack of publicly available datasets. Most research papers do not share their datasets, making it difficult to compare the effectiveness of different algorithms. Additionally, many studies inject artificial anomalies without explaining their methodology, leading to inconsistencies when benchmarking new models. Even when datasets are available, every company has its own internal policies that define abnormal behavior, and these policies can change over time due to evolving technical regulations or personnel decisions. Additionally, we observed that while models like Logistic Regression and SVM are easier to interpret, their lower recall rates make them less suitable for applications where missing a fault could have severe consequences. These models often struggle to capture complex relationships in data, particularly when anomalies are rare and subtle. Since recall measures the ability to identify all relevant anomalies, lower recall rates can lead to undetected faults, increasing the risk of operational failures. In high-stakes scenarios like fuel transportation, missing an anomaly could mean overlooking a potential safety hazard or a security threat, which could have severe financial and environmental consequences.

B. Analysis of Case Studies and Interpretation of Results

The results of this study highlight the potential of machine learning models to enhance fault and anomaly detection in fuel transportation systems. Among the models evaluated, XGBoost showed the greatest promise, with its superior performance across multiple metrics. The hybrid model using K-means and the isolation forest showed an accuracy of 0.95, and the other hybrid model combining supervised and unsupervised learning using K-Nearest Neighbor and the local outlier factor showed an accuracy of 0.80 and the two Variational Autoencoder models showed an average accuracy of 0.69 at the beginning and going up to 0.72 after incorporating Transfer Learning.

Another critical constraint is the context dependence of petroleum distribution anomalies. All anomalies picked up may not indicate faults or fraud; some deviations could be due to operational constraints, such as emergency redirection, fluctuating demand, or regulatory adjustments. Traditional machine learning models lack context awareness and hence cannot distinguish between genuinely alarming anomalies and permissible deviations, leading to potential false alarms. This could produce undesirable intervention during operations or causative delays. To negate this, the incorporation of future work could use external context knowledge such as weather, real-time logistic constraints, or domain knowledge regulations to strengthen anomaly interpretation and suppress false alarms.Additionally, hybrid approaches combining machine learning with expert-driven anomaly validation could enhance decision-making reliability.

This suggests that ensemble methods, which combine the strengths of multiple decision trees, are particularly effective for this type of problem. However, through this study we were also faced with limitations. The relatively small size of the dataset probably restricted the ability of our models to fully generalize, particularly for more complex scenarios. This limitation suggests that increasing the amount of data, particularly with more examples of faulty instances, could improve model performance.

Table I summarizes model performances, while Figure 2 shows the anomalies found in the territory of Italy divided by two clusters along with their centroids, where the red zone

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depicts the anomalies found in northern Italy and the green one in the central and southern parts.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.77	0.83	0.40	0.74
SVM	0.77	0.83	0.40	0.74
Random Forest	0.85	0.62	0.60	0.52
XGBoost	0.69	0.73	0.80	0.70
VAE	0.72	0.37	0.74	0.49
KMeans & IF	0.95	0.80	0.89	0.84
KNN & LOF	0.84	0.60	0.83	0.70

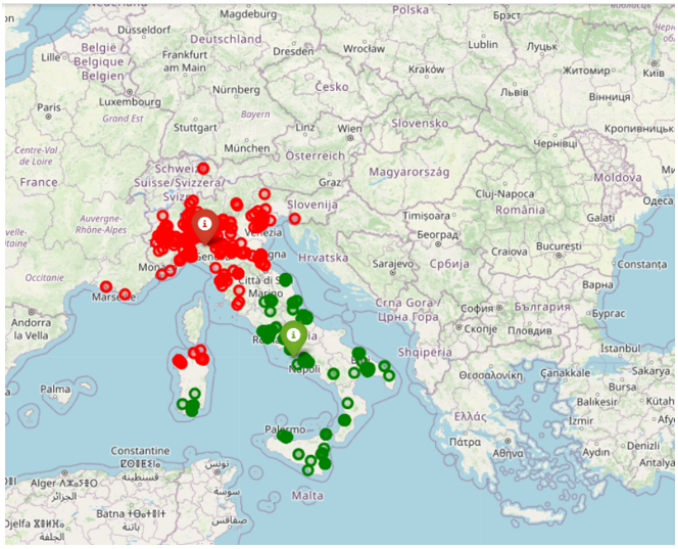

Fig. 2. Anomalies detected in the territory of Italy: red for northern zone, green for central and southern zones.

V. CONCLUSION AND FUTURE WORK

This paper presents a hybrid machine learning framework that integrates fault detection and anomaly detection in petroleum distribution. The findings indicate that supervised machine learning models effectively detect known equipment faults and unsupervised machine learning techniques detect unexpected anomalies in distribution. Combining both approaches enhances monitoring capabilities and operational safety.

Future work will explore how machine learning models can be deployed in real-time with IoT sensor integration to deep learning-based anomaly detection for more complex operational insights.

Automation and deployment will involve real-world monitoring of anomalies and faults using models, notification services, and stack logging systems. Other techniques such as SHAP, LIME, and feature importance analysis can enhance interpretive decisions.

Robustness and validation checks will help to assess the resilience of the trained model, test it using datasets from several domains or under various conditions and examine the model's capacity to identify adversarial examples deliberately manipulated data meant to trick the algorithm and see if it is capable of detecting faults and anomalies during fuel distribution.

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